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Calculus 1M - 104018

Midterm
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SOLUTION

1. (a) Let $\{a_n\}$ be a sequence and $L \in \mathbb{R}$. Provide the definition of $\lim_{n \rightarrow \infty} a_n = L$. (3 p)
- (b) Let $a_n = \frac{1}{n(n+2)}$. Prove by definition that $\lim_{n \rightarrow \infty} a_n = 0$. (7 p)

Answer

(a) For any $\epsilon > 0$ there exists $N = N(\epsilon) > 0$ such that $n > N$ implies $|a_n - L| < \epsilon$.

(b) $\lim_{n \rightarrow \infty} \frac{1}{n(n+2)} = 0$.

Proof: For any $\epsilon > 0$, take $N = \epsilon^{-1}$. Then

$$n > N = \epsilon^{-1} \implies \left| \frac{1}{n(n+2)} - 0 \right| = \frac{1}{n(n+2)} < \frac{1}{n} < \epsilon.$$

2. (a) Let $\{a_n\}$ be a sequence. Provide the definition of increasing sequence. (3 p)
 (b) Give an example of a non-increasing non-decreasing sequence. (5 p)
 (c) Is $a_n = \frac{n^2-n+2}{n}$ increasing ? If yes, prove your answer. If not, explain. (5 p)

Answer

- (a) The sequence is increasing if $a_n \leq a_{n+1}$ for any $n \in \mathbb{N}$.
 (b) The sequence $a_n = (-1)^n$ is non-increasing non-decreasing.
 (c) The sequence $a_n = \frac{n^2-n+2}{n}$ is increasing because:

$$\begin{aligned}
 a_n \leq a_{n+1} &\iff \frac{n^2 - n + 2}{n} \leq \frac{(n+1)^2 - (n+1) + 2}{n+1} \\
 &\iff (n+1)(n^2 - n + 2) \leq n((n+1)^2 - (n+1) + 2) \\
 &\iff n^3 - n^2 + 2n + n^2 - n + 2 \leq n(n^2 + 2n + 1 - n + 1) \\
 &\iff n^3 + n + 2 \leq n^3 + n^2 + 2n \\
 &\iff 2 \leq n^2 + n
 \end{aligned}$$

and the last inequality is true for any $n \in \mathbb{N}$.

3. (a) Give, if possible, an example of a monotonic sequence that doesn't have finite limit. (5 p)
- (b) Give, if possible, an example of a monotonic sequence that has finite limit. (5 p)

Answer

- (a) The sequence $a_n = n$ is monotonic increasing and it doesn't have finite limit.
- (b) The sequence $a_n = \frac{1}{n}$ is monotonic decreasing and its limit is 0.

4. Let $\{a_n\}$ be the sequence defined by

$$\begin{cases} a_1 = 2 \\ a_{n+1} = \sqrt{a_n + 2}. \end{cases}$$

Prove that

- (a) This sequence is bounded. (5 p)
- (b) This sequence is increasing. (5 p)
- (c) Find, if it exists, its limit. (5 p)

Answer

By induction we can prove that $a_n = 2$ for any $n \in \mathbb{N}$ since: $a_1 = 2$ and, if we assume that $a_n = 2$ for some $n \in \mathbb{N}$, then $a_{n+1} = \sqrt{a_n + 2} = \sqrt{2 + 2} = 2$.

The sequence is bounded since $2 \leq a_n \leq 2$ for any $n \in \mathbb{N}$.

The sequence is increasing since $a_n \leq a_{n+1}$ for any $n \in \mathbb{N}$.

Finally, $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} 2 = 2$.

5. (a) Provide the definition of injective function. (3 p)
- (b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = 1 + \sqrt{x^2 - 4}$.
- i. Find the domain of f . Justify.(3 p)
 - ii. Is f an injective function? Justify. (3 p)
 - iii. Find the image of f . Justify.(3 p)

Answer

- (a) A function f is injective if, for any $x_1, x_2 \in \text{Dom } f$,

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

- (b) i. $\text{Dom } f = \{x \in \mathbb{R} : x^2 - 4 \geq 0\} = \{x \in \mathbb{R} : |x| \geq 2\} = (-\infty, -2] \cup [2, +\infty)$.
- ii. The function f is not injective because $f(2) = f(-2)$ and $-2, 2 \in \text{Dom } f$.
- iii. Since $\sqrt{x^2 - 4} \geq 0$ then $f(x) = 1 + \sqrt{x^2 - 4} \geq 1$, so $\text{Im } f \subseteq [1, +\infty)$.
On the other hand, if $b \in [1, +\infty)$ then $b - 1 \geq 0$ and hence

$$b - 1 = \sqrt{(b - 1)^2} = \sqrt{a^2 - 4}$$

for $a = \sqrt{(b - 1)^2 + 4}$, with $a \geq 2$. Then $b = 1 + \sqrt{a^2 - 4} = f(a) \in \text{Im } f$ because $a \in \text{Dom } f$, and hence $[1, +\infty) \subseteq \text{Im } f$.

6. (a) Provide the definition of inverse function. (3 p)
- (b) Determine if the inverse exists. If yes, find its inverse function. If not, is it possible to change the domain and codomain in order to make it a bijection? If yes, do so and find the inverse function of f . If not, explain.
- i. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt[3]{x+4} - 2$; (3 p)
- ii. $f : (-1, 1) \rightarrow \mathbb{R}, f(x) = \frac{1}{1-x^2}$. (4 p)

Answer

- (a) The inverse of a function $f : A \rightarrow B$ is a function $f^{-1} : B \rightarrow A$ such that $(f \circ f^{-1})(x) = x$ for any $x \in B$, and $(f^{-1} \circ f)(x) = x$ for any $x \in A$.
- (b) i. The function $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt[3]{x+4} - 2$ is injective and surjective, hence it admits inverse, and $f^{-1}(x) = (x+2)^3 - 4$.
- ii. The function $f : (-1, 1) \rightarrow \mathbb{R}, f(x) = \frac{1}{1-x^2}$ is neither injective (since $f(-\frac{1}{2}) = f(\frac{1}{2})$) nor surjective (since $\text{Im } f = [1, +\infty)$). Hence its inverse does not exist.
- However, the functions $f_1 : [0, 1) \rightarrow [1, +\infty)$ and $f_2 : (-1, 0] \rightarrow [1, +\infty)$ admit inverse

$$f_1^{-1}(x) = \sqrt{1 - \frac{1}{x}}, \quad f_2^{-1}(x) = -\sqrt{1 - \frac{1}{x}}.$$

7. Fill the appropriate empty square with the letter **F** if it is False and the letter **T** if it is true. Prove true statements and provide counterexamples for false ones.
- (a) If the addition of two sequences $\{a_n + b_n\}_{n \geq 1}$ converges, then both sequences $\{a_n\}_{n \geq 1}$ and $\{b_n\}_{n \geq 1}$ converge.
 - (b) If two sequences $\{a_n\}_{n \geq 1}$ and $\{b_n\}_{n \geq 1}$ have limit $+\infty$, then the sequence $\{a_n - b_n\}_{n \geq 1}$ always converges to 0.
 - (c) If the addition of two sequences $\{a_n + b_n\}_{n \geq 1}$ and the sequence $\{a_n\}_{n \geq 1}$ converge, then the sequence $\{b_n\}_{n \geq 1}$ converges.

Answer

(a) FALSE:

Take $a_n = n$ and $b_n = -n$. Then $a_n + b_n = 0$ converges but a_n and b_n diverge.

(b) FALSE:

Take $a_n = 2n$ and $b_n = n$. Then $a_n - b_n = n$ has limit $+\infty$.

(c) TRUE:

If $\lim_{n \rightarrow \infty} (a_n + b_n) = L$ and $\lim_{n \rightarrow \infty} a_n = L'$ then

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} (a_n + b_n - a_n) = \lim_{n \rightarrow \infty} (a_n + b_n) - \lim_{n \rightarrow \infty} a_n = L - L'.$$

8. For each one of the following limits, find its value and indicate which properties of limits you are using to calculate it:

(a) $\lim_{n \rightarrow \infty} n^2 - n$ (5 p)

(b) $\lim_{n \rightarrow \infty} \sqrt[n]{3^n + 2}$ (5 p)

(c) $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n+2}\right)^{n+2}$ (5 p)

Answer

1. $\lim_{n \rightarrow \infty} n^2 - n = \lim_{n \rightarrow \infty} n(n-1) = +\infty$ since $\lim_{n \rightarrow \infty} n = \lim_{n \rightarrow \infty} n - 1 = +\infty$.

2. Observe first that $3^n \leq 3^n + 2 \leq 3^n + 3^n$ for any $n \in \mathbb{N}$. Then

$$3 = \sqrt[n]{3^n} \leq \sqrt[n]{3^n + 2} \leq \sqrt[n]{2 \cdot 3^n} = 3 \sqrt[n]{2}.$$

Since $\lim_{n \rightarrow \infty} \sqrt[n]{2} = 1$, the squeeze theorem implies that $\lim_{n \rightarrow \infty} \sqrt[n]{3^n + 2} = 3$.

3. $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n+2}\right)^{n+2} = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{n+2}\right)^{-(n+2)}\right]^{-1} = e^{-1}$.